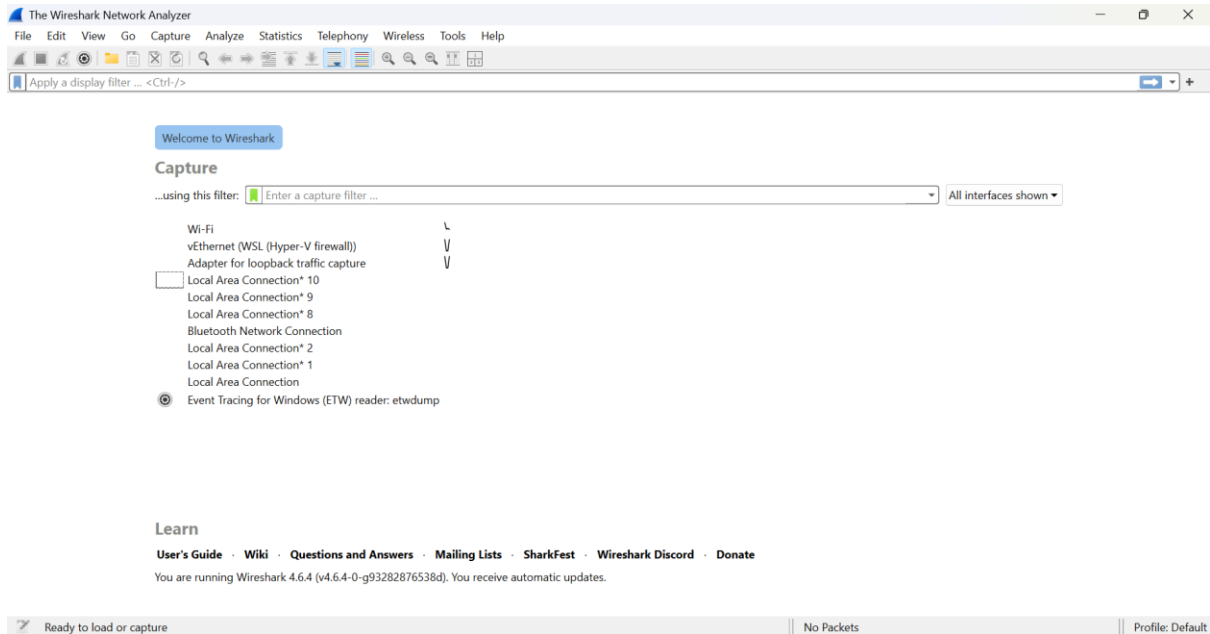


Assignment 1

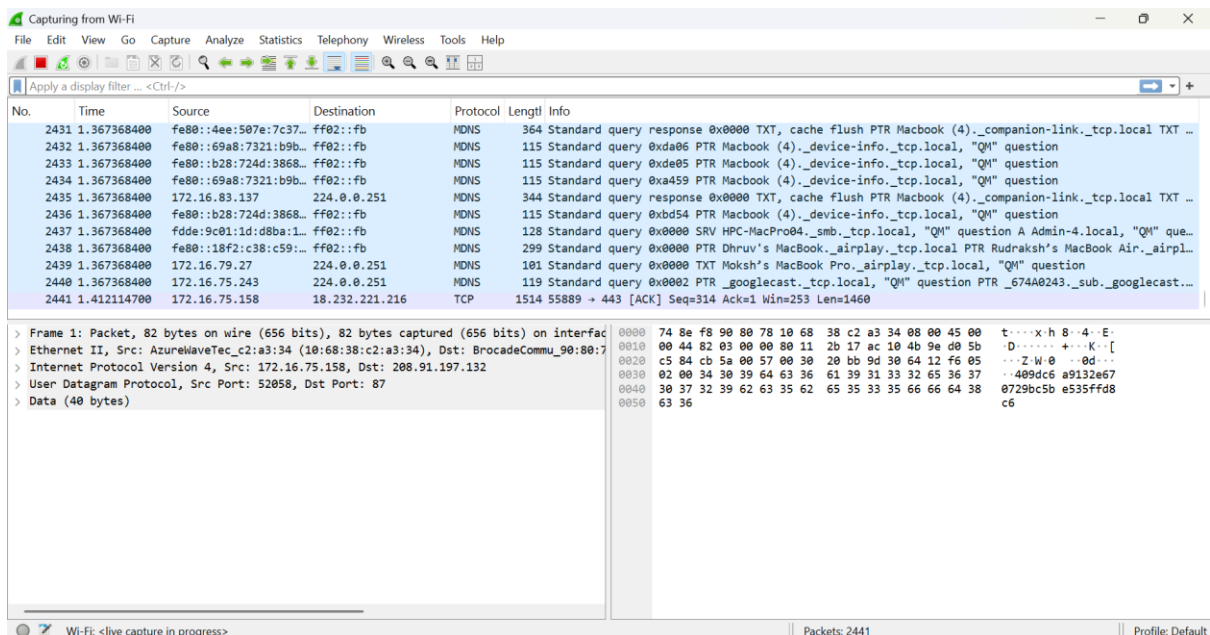
To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, DNS, and ICMP.

Step 1: Launch Wireshark



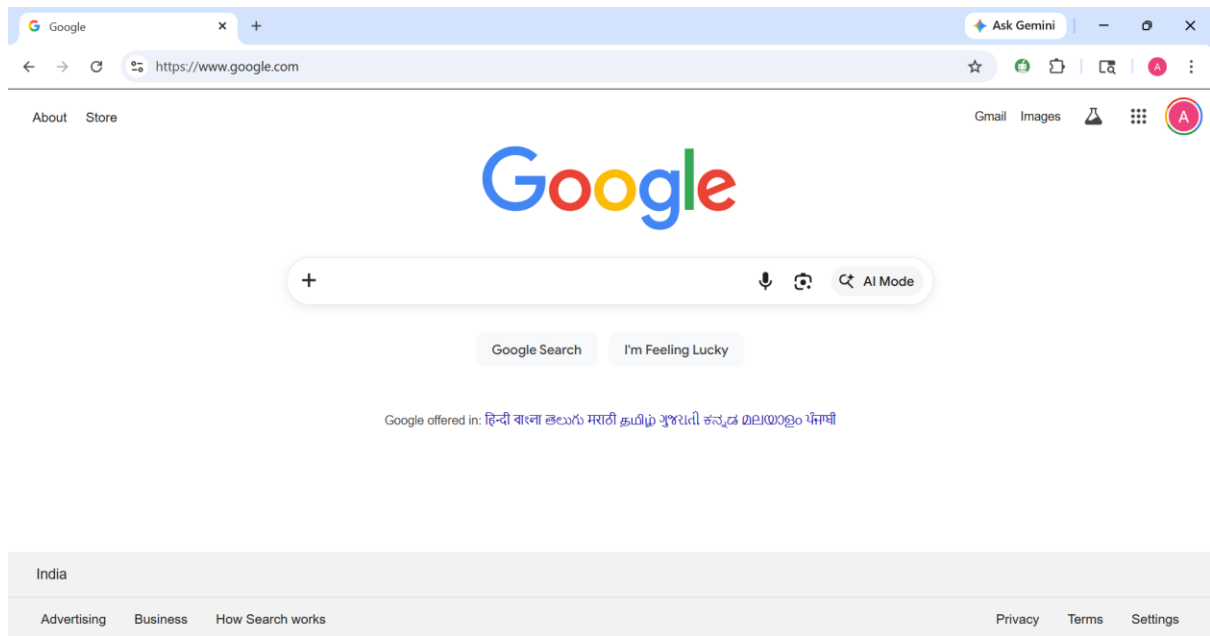
Step 2: Start Packet Capture

The active network interface (Wi-Fi) was selected, and packet capturing started.



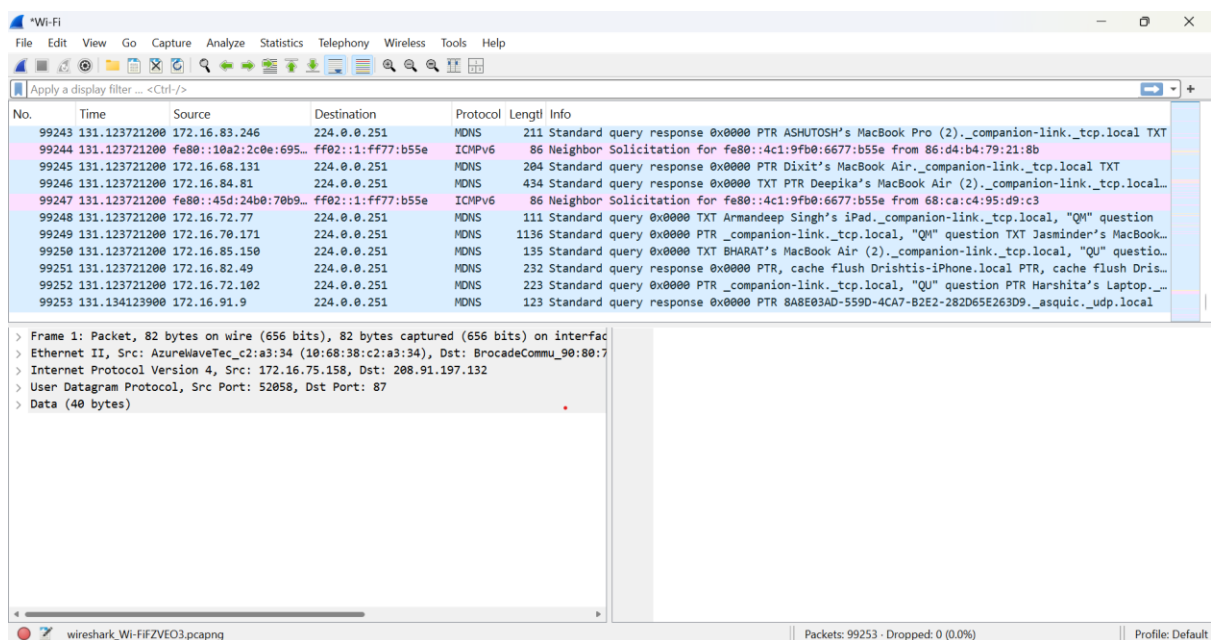
Step 3: Generate Network Traffic

To generate traffic: Opened a web browser and visited google.com



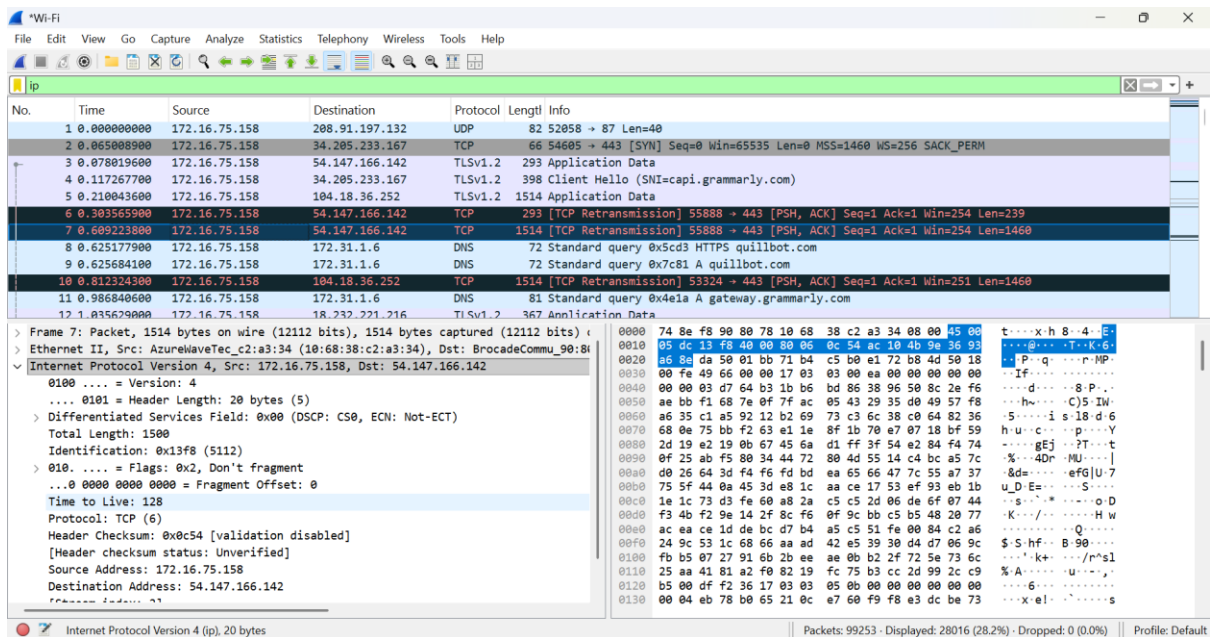
Step 4: Stop Capture

The packet capture was stopped using the red stop button in Wireshark.



Step 5: Apply Filter

Ip Filter Analysis



A packet was selected and analyzed using Wireshark. The following details were observed:

- **Ethernet Header:**
 - Shows source MAC address (AzureWaveTec device) and destination MAC address (network device).
- **IP Header:**
 - Version: IPv4
 - Source IP Address: 172.16.75.158
 - Destination IP Address: 54.147.166.142
 - Time to Live (TTL): 128
 - Protocol: TCP
- **TCP Header:**
 - Source Port: 55888
 - Destination Port: 443 (HTTPS)
 - Flags: PSH, ACK
 - Indicates data transmission over a secure connection
- **Payload Data:**
 - Contains encrypted application data (since HTTPS is used)

TCP Filter Analysis

The image shows a Wireshark packet capture window titled "Wi-Fi". The packet list on the left shows a series of TCP and TLSv1.2 packets. Packet 10 is selected, showing details for the Transmission Control Protocol (TCP). The packet is from source port 53324 to destination port 443, with sequence number 1 and acknowledgment number 1. The flags are PSH and ACK. The window size is 251. The packet length is 1460 bytes. The raw data is shown in hexadecimal and ASCII format on the right.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.065008900	172.16.75.158	34.205.233.167	TCP	66	54605 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3	0.078019600	172.16.75.158	34.205.233.167	TLSv1.2	293	Application Data
4	0.117267700	172.16.75.158	34.205.233.167	TLSv1.2	398	Client Hello (SNI=capi.grammarly.com)
5	0.210043600	172.16.75.158	104.18.36.252	TLSv1.2	1514	Application Data
6	0.303565900	172.16.75.158	54.147.166.142	TCP	293	[TCP Retransmission] 55888 → 443 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=239
7	0.609223800	172.16.75.158	54.147.166.142	TCP	1514	[TCP Retransmission] 55888 → 443 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=1460
10	0.812324300	172.16.75.158	104.18.36.252	TCP	1514	[TCP Retransmission] 53324 → 443 [PSH, ACK] Seq=1 Ack=1 Win=251 Len=1460
12	1.035629000	172.16.75.158	18.232.221.216	TLSv1.2	367	Application Data
13	1.066817700	172.16.75.158	34.205.233.167	TCP	66	[TCP Retransmission] 54605 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14	1.223327500	172.16.75.158	54.147.166.142	TCP	1514	[TCP Retransmission] 55888 → 443 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=1460
15	1.270488400	172.16.75.158	18.232.221.216	TCP	367	[TCP Retransmission] 55889 → 443 [PSH, ACK] Seq=1 Ack=1 Win=253 Len=313

Transmission Control Protocol, Src Port: 53324, Dst Port: 443, Seq: 1, Ack: 1, Len: 1460

[Stream index: 3]
Source Port: 53324
Destination Port: 443
[Stream index: 3]
[Stream Packet Number: 2]
> [Conversation completeness: Incomplete (28)]
[TCP Segment Len: 1460]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3382901911
[Next Sequence Number: 1461 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment Number (raw): 3406699494
001 ... = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 251

0020 24 fc d0 4c 01 bb c9 a2 fc 97 cb 0e 1b e6 18 \$.L.....
0030 00 fb 3f a1 00 00 17 03 03 00 ef eb ab cd eb 2a ..?.....*
0040 81 10 9c da 39 30 cf 28 81 46 02 fe db 3b e1 9b ...90((F...;
0050 45 fb c2 7c 31 35 cc 97 45 91 52 9e bd d4 a6 6c E..|15- E-R...l
0060 c1 d0 c2 94 3d 9c 3c 53 aa 48 1e e0 97 d0 2d b7<S H.....
0070 f5 cf ab 14 35 34 74 a7 6f a5 58 f9 4b c3 11 35 ...54t. o X K > 5
0080 79 3a 1e f1 94 df 44 ad 4d cc 8d e1 d0 e6 41 cd y:....DM M....A
0090 48 6c 10 90 ac c5 dd aa 18 c5 e7 c8 64 80 96 f2 H1.....<...d
00a0 e5 0f a6 81 17 86 1b 38 3b c0 47 b1 db 31 49 a78 ; G-1I-
00b0 f2 9e 71 7d bf af 69 c7 3c 12 49 e0 85 94 64 bb ...q}...i < I-...d
00c0 9b 60 10 3c 3f 85 c5 5d 25 ef 95 48 60 78 34 38 ...<?..] %-H"x48
00d0 51 c7 03 d1 18 ad 32 57 63 5c 87 05 a4 7a 03 06 Q.....2W c\...z-
00e0 4c 1c 1c 56 77 49 42 6c 42 fe c3 1a 7d ab cd 61 L..VwIB1 B...}a
00f0 53 44 76 0a d5 67 c5 6e 4e 01 01 3e ec 7e 82 2b SDV. g n N->...+
0100 5b c0 34 76 75 f4 d6 a3 d6 9a 5e 32 2a d8 e8 23 [.4vu.....?2*...#
0110 90 5c 62 fb 1b b8 11 d0 d6 ce 43 f9 8c 1a 7c ed \b.....C...]-
0120 07 49 38 36 67 49 38 0c 0b 99 17 03 03 0f 64 24 .I86gI8.....d\$
0130 c7 79 de 7d f1 29 4b 55 4b d6 3f 9d 2e 7a 6c 3b .y-})KU K?..z1;
0140 15 49 83 5f 00 3b 97 fc 7f 49 da 3e 1b 7d 22 54 .I.....I->}T
0150 3c 12 09 76 1e a0 da 05 dc 2a 7d 55 04 d5 a9 5e <-V.....*)U...^

TCP packets show connection-oriented communication. The presence of:

- SYN flag indicates connection establishment
- ACK flag confirms receipt of data
- PSH flag indicates immediate data transfer

Detailed TCP segment analysis shows:

- Sequence Number: 1
- Acknowledgment Number: 1
- Header Length: 20 bytes
- Window Size: 251
- Flags: PSH, ACK

DNS Filter Analysis

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
8	0.356058500	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	94	Standard query 0x5220 A assets.msn.com
9	0.356108100	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	94	Standard query 0xc62c AAAA assets.msn.com
12	0.389332000	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	318	Standard query response 0xc62c AAAA assets.msn.com CNAME assets-msn-com-world-atm-default.traff...
13	0.381800000	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	266	Standard query response 0x5220 A assets.msn.com CNAME assets-msn-com-world-atm-default.traff...
287	24.4087851700	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	94	Standard query 0x75e9 A www.google.com
288	24.408841500	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	94	Standard query 0x37b7 AAAA www.google.com
289	24.621656200	192.168.29.212	192.168.29.1	DNS	74	Standard query 0x75e9 AAAA www.google.com
290	24.622110300	192.168.29.212	192.168.29.1	DNS	74	Standard query 0x75e9 A www.google.com
291	24.633681300	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	222	Standard query response 0x75e9 A www.google.com A 142.251.156.119 A 142.251.150.119 A 142.251.1...
292	24.633681300	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	318	Standard query response 0x37b7 AAAA www.google.com AAAA 2001:4860:4829:7700:1:AAAA 2001:4860:48...
297	25.124979500	192.168.29.1	192.168.29.212	DNS	202	Standard query response 0x75e9 A www.google.com A 142.251.153.119 A 142.251.156.119 A 142.251.1...
469	37.213407300	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	100	Standard query 0xab05 A beacons.gcp.gvt2.com
476	37.213640300	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	100	Standard query 0xd2ac AAAA beacons.gcp.gvt2.com
478	37.216077800	2405:201:5019:5807::...	2405:201:5019:5807::...	DNS	111	Standard query 0x055b AAAA signaler-pa.clients6.google.com

```

> Frame 8: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on inter-
> Ethernet II, Src: AzuleviaTec-c2:a3:34 (10:68:38:c2:a3:34), Dst: SkyworthDigi_61:4
> Internet Protocol Version 6, Src: 2405:201:5019:5807:45b1:c91f:86d5:1c79, Dst: 2405
0110 ... = Version: 6
  > .... 0000 0000 ... = Traffic Class: 0x00 (DSCP: CS0, ECN: Nc
    .... 0000 00 ... = Differentiated Services Codepoint: D
      .... 00 ... = Explicit Congestion Notification: Nc
    .... 0010 1001 1110 0111 1011 = Flow Label: 0x29e7b
    Payload Length: 40
    Next Header: UDP (17)
    Hop Limit: 64
  > Source Address: 2405:201:5019:5807:45b1:c91f:86d5:1c79
  > Destination Address: 2405:201:5019:5807::c0a8:1001
    [Stream index: 0]
  
```

Domain Name System: Protocol

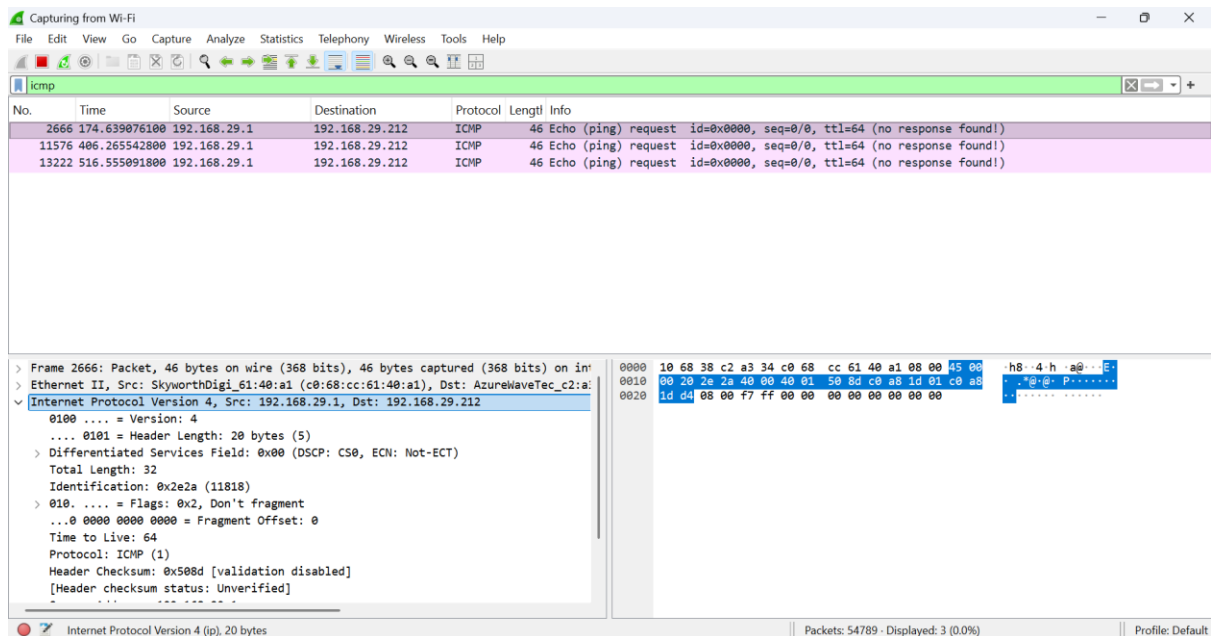
Packets: 49573 - Displayed: 901 (1.8%)

Profile: Default

The dns filter was applied to display Domain Name System packets. DNS is responsible for translating domain names into IP addresses. The captured packets show DNS queries and responses between the client and DNS servers.

- Protocol: DNS
- IP Version: IPv6
- Source Address: 2405:201:5019:5807:... (your system)
- Destination Address: DNS server (same IPv6 range)
- Query Type: A (IPv4 address), AAAA (IPv6 address)
- Domains Queried:
 - assets.msn.com
 - www.google.com
 - beacon.gcp.gvt2.com

ICMP Filter Analysis:



The icmp filter was applied to display Internet Control Message Protocol packets. ICMP is used for diagnostic and error-reporting purposes, such as the ping command.

- Protocol: ICMP
- Source IP Address: 192.168.29.1
- Destination IP Address: 192.168.29.212
- Message Type: Echo (ping) request
- TTL (Time to Live): 64